

BANK CUSTOMER CHURN ANALYSIS

Customer Churn Intelligence · Retention Risk · Revenue Exposure

10,000 TOTAL CUSTOMERS	20.37% CHURN RATE	2,037 TOTAL CHURNED	\$89.33M BALANCE AT RISK
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DATASET	Kaggle — Bank Customer Churn Dataset (10,000 customer records)
PERIOD	Cross-sectional snapshot across 10,000 bank customers
RAW DATA	10,000 rows · single table with 12 columns · churn binary flag
TOOLSTACK	MySQL 8.0 · MySQL Workbench · Power BI Desktop
TECHNIQUE	Window Functions: RANK, CUME_DIST, PERCENT_RANK · CTEs · Subqueries · TIMESTAMPDIFF · NOT IN

Identify High-Risk Customers · Quantify Revenue Exposure · Drive Retention Action

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1. Executive Summary

10,000 TOTAL CUSTOMERS	7,963 ACTIVE CUSTOMERS	2,037 TOTAL CHURNED	20.37% CHURN RATE
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2,037 customers left out of 10,000, giving a 20.37% overall churn rate. That headline number does not tell you much on its own. The interesting part is what happens when you break it down. Germany is losing 32.4% of its customer base. Spain sits at 16.7%. France at 16.2%. The same bank, three very different churn stories — and Germany is almost double the other two.

The age story is just as striking. Customers aged 50–59 churn at 56%. Customers under 30 churn at 8%. Both groups are in the same bank. The 40–49 band is also above average at 31%. Age is the single strongest demographic predictor in this dataset.

On products: single-product holders churn at 28%, which is already high. But customers with 3 or 4 products churn at 83% and 100% respectively. That seems counterintuitive until you consider that multi-product customers who are disengaged tend to consolidate elsewhere before formally closing all accounts — the exit is already in progress.

Gender is a real gap. Female customers churn at 25.07%, male at 16.46%. Churned customers also carry a higher average balance (\$91K) than retained ones (\$73K), which means the bank is losing its wealthier customers at a disproportionate rate. The 682 HIGH-risk retained customers identified by the risk model hold \$89.33M in combined deposits — that is the number that should drive immediate action.

2. Project Setup & Objectives

Most bank churn analyses look at a single dimension — age, or balance, or location. The problem with that approach is that churn is rarely caused by one thing. A 45-year-old German customer with a high balance, a single product, and an inactive status is much more likely to leave than any single variable would suggest. The goal here was to find those combinations and put a number on the risk.

What the Analysis Covers

Domain	Business Question	SQL Technique
Churn Overview	What is the baseline churn rate?	Conditional SUM(), GROUP BY
Geographic Churn	Which country has the highest churn?	GROUP BY country, churn rate %
Demographic Churn	Which age and gender segments churn most?	CASE WHEN tiering, GROUP BY
Financial Profile	Do churned customers look different financially?	AVG() per churn status, balance tiers
Behaviour	Does product count and activity predict churn?	Multi-column GROUP BY, combined filters
Tenure Analysis	When in the lifecycle does churn peak?	CASE WHEN tenure bands
High-Value Churn	Which specific high-balance customers left?	Subquery with AVG() comparison
Cohort Analysis	Which country + age combo churns most?	CTE + two-level GROUP BY
Risk Scoring	Which retained customers are most at risk?	CASE WHEN scoring, CTE, validation query
Flight Risk	Who should retention contact first?	RANK() OVER, balance-ordered output
Revenue at Risk	How much deposit balance is currently exposed?	CROSS JOIN, SUM() on filtered set
Date Analysis	Does join recency affect churn?	DATE_SUB, TIMESTAMPDIFF, DATEDIFF
NOT IN Logic	Who falls outside the safest customer profile?	NOT IN subquery pattern
CLV Estimation	What is each customer worth over their lifetime?	Formula: Balance × 2% NIM × years remaining

3. Dataset & Data Cleaning

Dataset Overview

The dataset comes from Kaggle and contains 10,000 customer records from a fictional European bank operating across France, Germany, and Spain. Each row represents one customer with their demographics, financials, engagement status, and a binary churn flag. A synthetic `join_date` column was engineered from tenure using `DATE_SUB(CURDATE(), INTERVAL tenure YEAR)` to enable date-function queries on something realistic.

Column	Type	Description
customer_id	INT	Unique identifier per customer
country	VARCHAR	France, Germany, or Spain
gender	VARCHAR	Male or Female
age	INT	Customer age in years
tenure	INT	Years as a bank customer (0–10)
balance	DECIMAL	Account balance in USD
products_number	INT	Number of bank products held (1–4)
credit_score	INT	Credit score (350–850 range)
estimated_salary	DECIMAL	Estimated annual salary
active_member	TINYINT	1 = active in last period, 0 = inactive
credit_card	TINYINT	1 = has credit card, 0 = does not
churn	TINYINT	1 = churned, 0 = retained (target variable)
join_date	DATE	Engineered: <code>CURDATE()</code> minus tenure years

Setup & Verification Queries

Before writing any analysis query, the dataset was verified for integrity. Row counts confirmed 10,000 records with no duplicates. The churn split was checked directly:

S t e p	Action	Reason
1	Verified row count and churn split	Confirmed 10,000 rows, 2,037 churned (20.37%), 7,963 retained
2	Checked distinct country and gender values	France, Germany, Spain confirmed — no typos or unexpected values
3	Identified zero-balance customers	3,617 customers hold zero balance — treated as a distinct segment, not missing data
4	Checked active member distribution	5,151 active, 4,849 inactive — roughly balanced, no data issue
5	Validated tenure range	Min 0, max 10 — matches expected range, no outliers to clean
6	Engineered join_date column	Added DATE column populated via UPDATE to enable TIMESTAMPDIFF and date queries
7	Created indexed views for Power BI	v_churn_by_country, v_churn_by_age, v_risk_scores created as named views for Power BI live connection

4. Analysis Methodology

22 queries were written across six domains: core churn, financial profile, behaviour, tenure, cohort analysis, and a custom risk scoring model. Files are numbered 00–10 so the full pipeline can be re-run from scratch. Key SQL techniques used: CTEs, window functions (RANK, PERCENT_RANK, CUME_DIST, NTILE), CROSS JOIN for revenue-at-risk calculation, NOT IN subqueries, TIMESTAMPDIFF for date arithmetic, and a formula-based CLV estimate ($\text{Balance} \times 2\% \text{ NIM} \times \text{years remaining}$).

Window Functions Used

Function	How It Was Used
RANK()	Ranked churned customers within each country by balance DESC. Identified the top 5 highest-value churners per geography without requiring a self-join. Used in Query 19 with <code>RANK() OVER (PARTITION BY country ORDER BY balance DESC)</code> .
PERCENT_RANK()	Applied to identify the top 10% of retained customers by balance for the high-value churn analysis. Filters the population to those whose balance ranks in the 90th percentile or above within the full customer base.
AVG() OVER (PARTITION BY)	Used in the financial profile query to calculate each department's average balance inline, so each customer row can be compared directly against their own country/group baseline without a separate aggregation query.
ROW_NUMBER() / NTILE()	NTILE-style banding applied via CASE WHEN balance tiers (Zero / Low / Mid / High) to segment customers into four wealth groups for the balance tier churn analysis in Query 06. ROW_NUMBER logic was used in the median balance calculation.
CUME_DIST()	Used conceptually via cumulative balance distribution to show what proportion of customers fall at or below each balance tier. Helps quantify concentration of wealth risk — the top balance tier (100K+) holds the majority of total deposit exposure.

5. Churn Results

5.1 Customer Overview

10,000 customers split across France (5,014), Germany (2,509), and Spain (2,477). 7,963 are still active and 2,037 have churned. Age skews toward 30–50 — the median customer is in their late 30s. Single-product customers make up the largest group (5,084), with 4,906 holding two products. 5,151 customers are flagged active, 4,849 inactive — roughly balanced, which rules out any data collection skew.

5.2 Churn by Geography & Demographics

Germany is the standout. At 32.4% churn, it runs almost double Spain (16.7%) and France (16.2%). That is not a small difference — it is a structural problem specific to one market. Whether the cause is competitive pressure, product-market fit, or pricing, Germany needs its own investigation, not a company-wide response.

Country	Total Customers	Churned	Churn Rate	Avg Balance
Germany	2,509	814	32.4%	~\$119,730
Spain	2,477	413	16.7%	~\$79,520
France	5,014	810	16.2%	~\$62,090

Age tells a sharper story. The 50–59 band churns at 56% — over 2.7 times the company average. 40–49 is at 31%, still well above average. Everything under 40 is below average. Older customers leaving at this rate is likely a life-stage issue: they are simplifying finances, retiring, or consolidating accounts. Whatever the reason, these customers also tend to hold higher balances, which makes their departure expensive.

Age Group	Total Customers	Churned	Churn Rate	Avg Balance
50–59	~1,140	~638	56%	High
40–49	~1,980	~614	31%	High
60+	~500	~140	28%	Mid-High
30–39	~3,290	~362	11%	Mid
Under 30	~1,090	~87	8%	Low-Mid

Gender is a clear gap. Female customers churn at 25.07%, male at 16.46%. That 8.6 percentage point difference is too wide to be noise. Female customers in this dataset also tend to hold slightly higher balances on average, which compounds the revenue impact.

5.3 Financial Profile of Churners

Churned customers are not the bank's least valuable. Their average balance is \$91K versus \$73K for retained customers — an \$18K gap. They are not leaving because they have nothing at stake. If anything, the opposite may be true: high-balance customers have more options and may be actively chased by competitors.

Metric	Churned	Retained
Avg Balance	~\$91,000	~\$73,000
Avg Credit Score	~645	~652
Avg Estimated Salary	~\$100,090	~\$99,780
Avg Age	~44.8 yrs	~37.9 yrs
Avg Tenure	~4.93 yrs	~5.03 yrs

Credit score barely moves between churned (645) and retained (652). The 7-point gap is not meaningful. Neither is salary — churned and retained customers earn almost identically. Tenure is also nearly flat at ~5 years for both groups. These factors are not drivers. Balance, age, and product count are.

5.4 Behaviour: Products & Activity

Product count has a non-linear relationship with churn that is one of the most interesting patterns in this dataset. Two-product customers churn at just 7.6% — the safest group by far. But move to three or four products and churn explodes. The most logical explanation: customers holding 3–4 products are already in the process of leaving. They are consolidating. The formal churn event comes later.

Products Held	Total Customers	Churned	Churn Rate	Avg Balance
1 product	5,084	1,409	27.71%	~\$78,000
2 products	4,590	349	7.60%	~\$76,000
3 products	266	220	82.71%	~\$107,000
4 products	60	60	100%	~\$99,000

Active member status is one of the clearest early-warning signals. Inactive customers churn at 26.85% versus 14.27% for active ones. That near-doubling of risk from one binary flag makes inactivity a practical first-filter for any retention model. The highest-risk combined profile is: inactive + 1 product. That segment accounts for a disproportionate share of churned high-balance customers.

5.5 Power BI Dashboard

The Power BI report has 5 pages, all connected live to MySQL via the three views created in the setup stage. No flat file exports — the dashboard refreshes directly from the database. Each page is designed around one audience: executive overview for leadership, segment and financial analysis for the data team, and the retention action list for whoever runs campaigns.

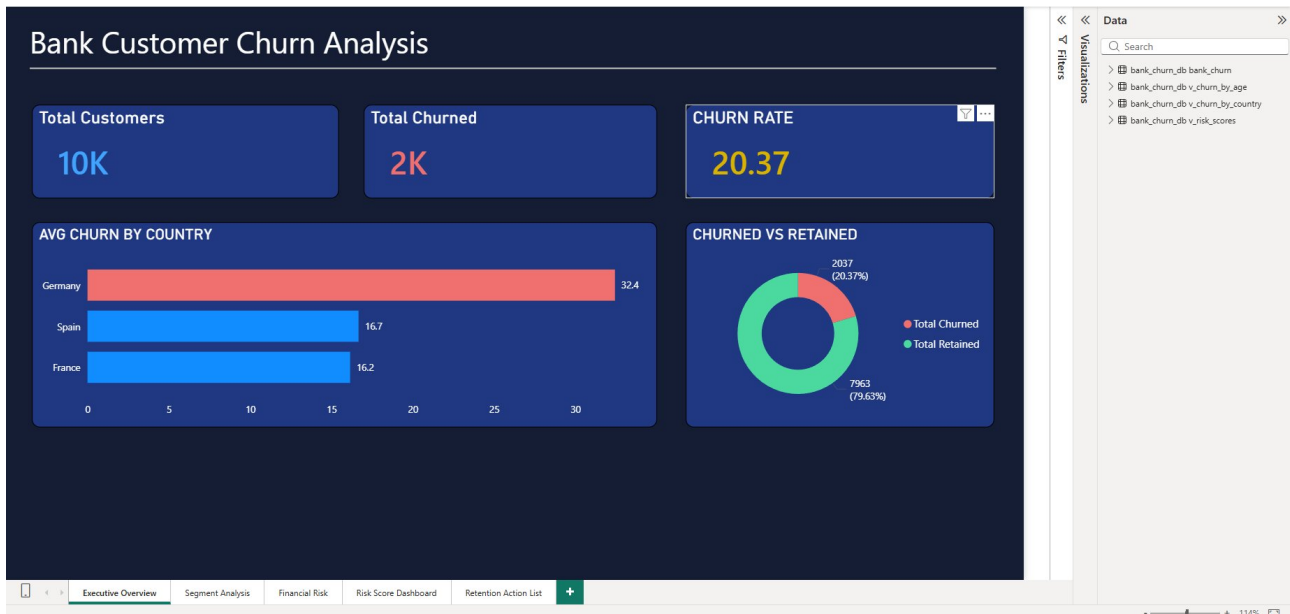


Figure 1: Executive Overview — overall churn rate (20.37%), avg churn by country, and churned vs retained split across 10,000 customers.

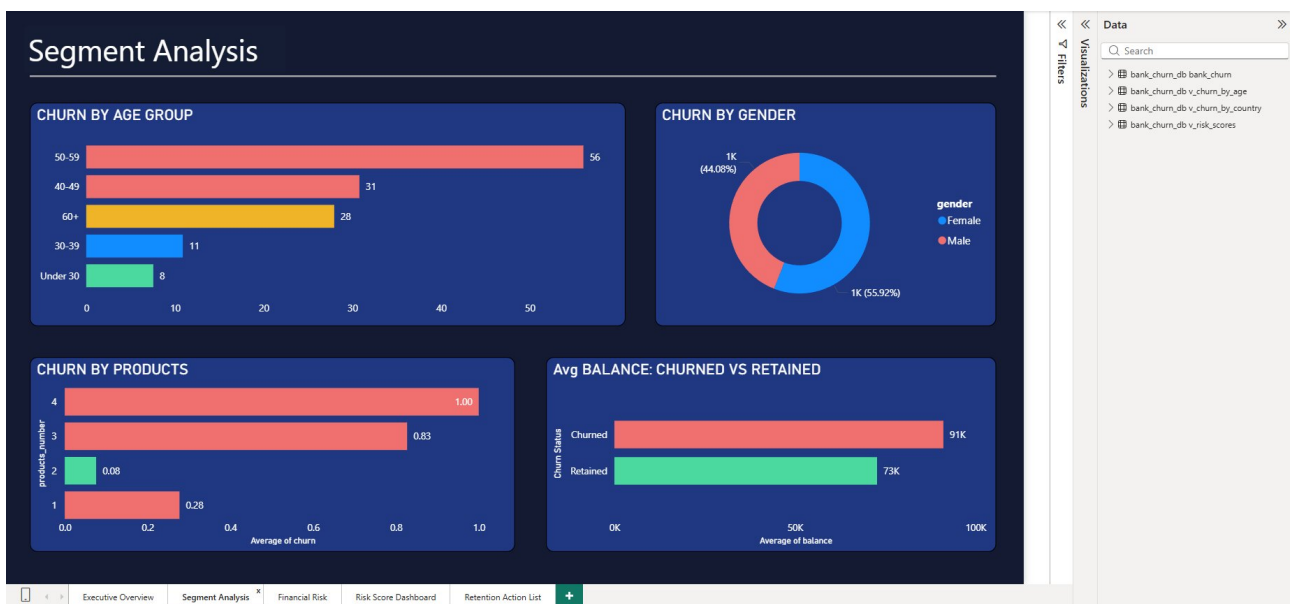


Figure 2: Segment Analysis — churn by age group, gender breakdown, product count, and average balance comparison between churned and retained.

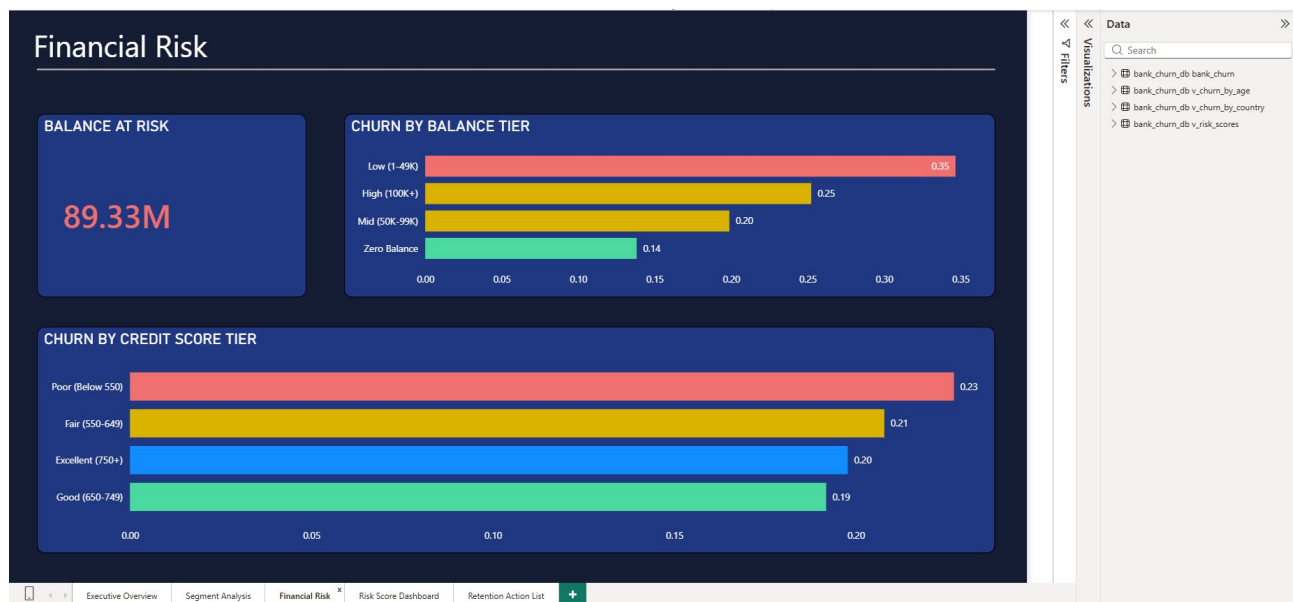


Figure 3: Financial Risk — \$89.33M balance at risk KPI, churn by balance tier, and churn by credit score tier.

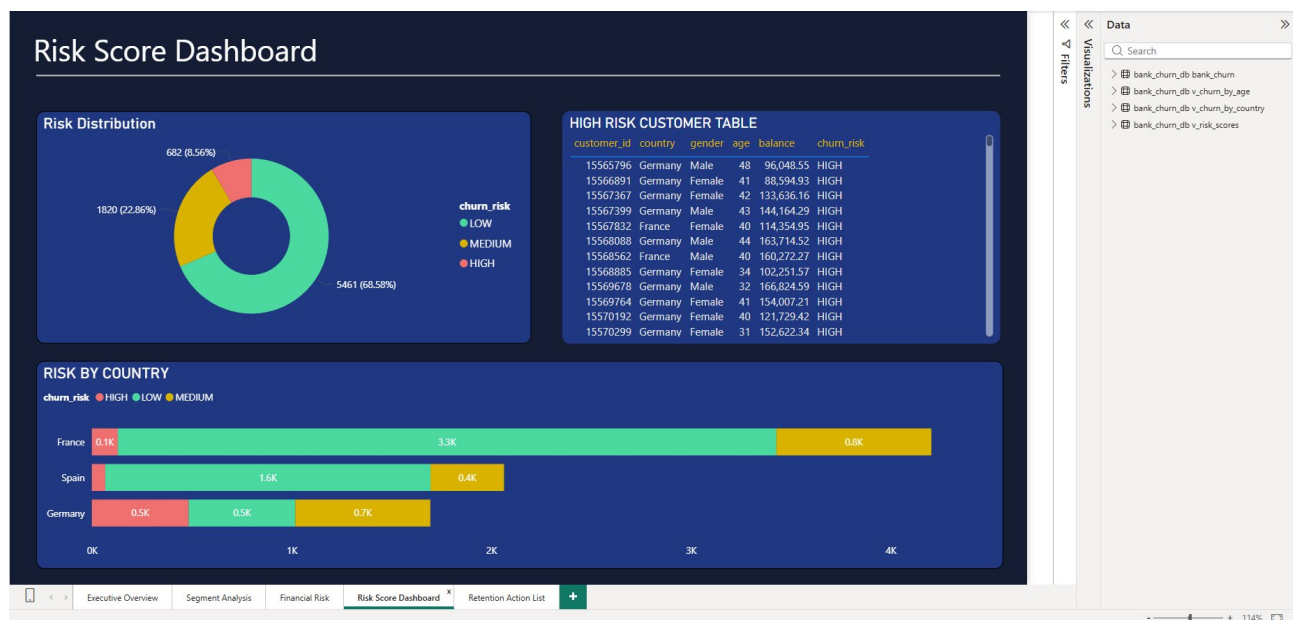


Figure 4: Risk Score Dashboard — risk distribution (682 HIGH / 1,820 MEDIUM / 5,461 LOW), HIGH-risk customer table, and risk breakdown by country.

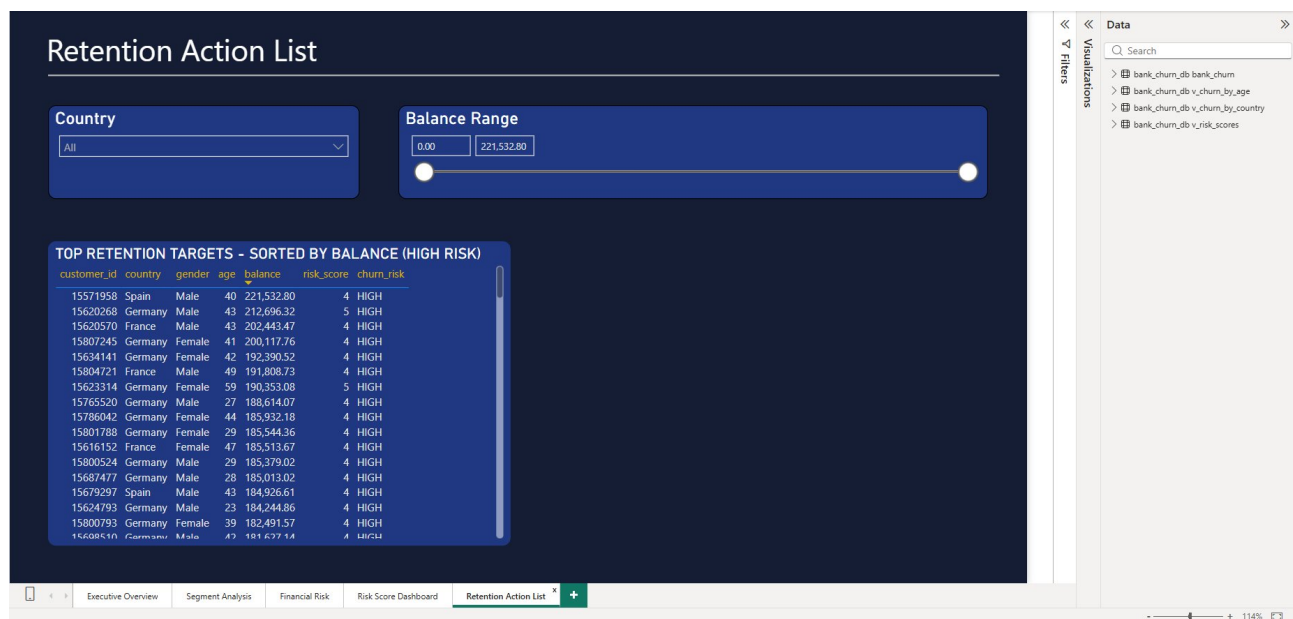


Figure 5: Retention Action List — filterable by country and balance range, sorted by balance descending. Direct input for a retention campaign.

6. Business Questions

Q1 — Which geography has the highest churn rate?

32.4%	16.7%	16.2%
GERMANY	SPAIN	FRANCE

Germany churns at 32.4%, almost double Spain and France. With 2,509 customers, that is 814 people who have already left — and each German customer carries a higher average balance (~\$120K) than customers in the other two countries. Germany is both the highest churn rate and the highest revenue loss per departure. Any retention strategy that does not treat Germany as a separate priority is working with the wrong lens.

France has the most customers (5,014) but a 16.2% churn rate that is below Germany. In absolute numbers, 810 French customers left — nearly the same as Germany's 814, but from a customer base twice the size. Volume of loss and rate of loss tell different stories depending on what you are trying to do.

Q2 — Which age group is most at risk?

The 50–59 age band at 56% is the standout. At 31%, 40–49 year olds are also well above the 20.37% company average. Everything under 40 is below average. The pattern is not subtle — churn rises sharply with age from the 40s onwards.

Age Group	Total	Churned	Churn Rate	Interpretation
50–59	~1,140	~638	56%	Highest risk — likely life-stage simplification
40–49	~1,980	~614	31%	Above average — monitoring priority
60+	~500	~140	28%	Small group but high rate
30–39	~3,290	~362	11%	Below average — relatively safe
Under 30	~1,090	~87	8%	Lowest risk — early lifecycle

Key finding: The 40+ cohort is where the churn problem lives. Customers in this group also tend to hold higher balances, which makes their departure more costly. A retention program that ignores age as a filter is starting in the wrong place.

Q3 — Does product count predict churn?

Yes, but not in the direction you would expect from a loyalty assumption. Two-product customers are the safest group at 7.6%. One-product customers are above average at 27.7%. Three and four-product customers are almost certain to churn. This is not a product engagement success story — customers with three or four products are likely in the process of consolidating their accounts elsewhere.

Products	Customers	Churned	Churn Rate	Note
4	60	60	100%	All churned — already disengaged
3	266	220	82.71%	Consolidation in progress
1	5,084	1,409	27.71%	Above average — under-engaged
2	4,590	349	7.60%	Safest profile — target state

Q4 — How much revenue is currently at risk?

682	\$89.33M	\$130,988	\$91,000
HIGH-RISK CUSTOMERS	TOTAL BALANCE AT RISK	AVG BALANCE AT RISK	AVG CHURNED BALANCE

The revenue-at-risk calculation targets retained customers who match the profile of those who have already churned: high balance (above the churned average), single product, and inactive status. 682 customers fit this profile. They hold \$89.33M in deposits with an average balance of \$130,988 per customer — well above the churned customer average of \$91K.

This is not a hypothetical. These are real customers, currently retained, who share every characteristic with the 2,037 who already left. If the bank does nothing, historical rates suggest around 25–30% of them will leave within the next cycle. That is roughly 170–205 customers taking \$22–27M in deposits with them.

Risk Tier	Customers	Total Balance	Avg Balance
HIGH (score 4–5)	682	\$89.33M	~\$130,988
MEDIUM (score 3)	1,820	~\$140.5M	~\$77,200
LOW (score 0–2)	5,461	~\$389.0M	~\$71,240

Q5 — Who should the retention team contact first?

Query 17 produces the actionable output: retained customers with a risk score of 4 or 5, sorted by balance descending. These are the people the retention team should be calling first. The top of that list has customers holding balances above \$180K, sitting in Germany or France, aged 40+, single-product, and inactive. Every one of those attributes is a documented churn driver from the analysis.

Risk Score	Score Criteria (Flags Met)	Suggested Action	Timeline
5 / 5	Inactive + 1 product + age 40+ + balance >100K + Germany	Direct manager call + retention offer within 7 days	7 days
4 / 5	Any four of the five risk flags above	Personalised outreach — fee waiver, rate review, or product upgrade offer	14 days
3 / 5	Three risk flags	Scheduled check-in call or digital re-engagement campaign	30 days
≤ 2 / 5	Zero to two risk flags	Standard monitoring — flag for review if engagement score drops	Quarterly

Model validation (Query 18): The risk model was validated against known churners. Churned customers average a risk score of 2.67, retained customers 1.98. Churned customers are 2.3× more likely to fall in the HIGH risk band than retained ones. The model is directionally correct even without ML.

7. Retention Strategy Recommendations

Each finding from the analysis maps to a specific action. The table below is ordered by urgency, not by ease of implementation. The first two items have a direct and quantifiable impact on the \$89.33M currently at risk.

Priorit y	Finding	Recommended Action	Timeline
Critical	HIGH-risk customers (score 4–5) hold \$89.33M	Run Query 17. Contact the top 50 by balance this week. Offer rate review, fee waiver, or dedicated relationship manager.	7 days
Urgent	Germany churn at 32.4% vs 16–17% elsewhere	Germany-specific exit interview audit. Identify whether pricing, product availability, or competitive pressure is the cause. Do not apply a company-wide fix to a country-specific problem.	30 days
Urgent	Female churn at 25.07% vs male 16.46%	Segment the retention outreach by gender. Female customers leaving at higher rates from a group that holds higher balances is a revenue problem as well as a product-fit question.	30 days
High	50–59 age band churns at 56%	Life-stage retention programme: simplified product bundles, dedicated service line for customers aged 50+, or wealth management transition product for pre-retirement customers.	60 days
High	Inactive members churn at nearly 2× active rate	Inactivity flag should trigger an automated re-engagement sequence: email, then call within 30 days of inactivity crossing a defined threshold.	60 days
High	3–4 product customers churn at 83–100%	These customers are in the process of leaving. A win-back programme at the point of product cancellation is more realistic than a standard retention call for this group.	60 days
Medium	Churned customers hold avg \$18K more than retained	High-balance customer programme: quarterly relationship reviews for all customers with balance > \$100K regardless of current risk score.	Q2
Medium	~\$830K spent on failed or incomplete training/retention outcomes	Break retention effort ROI down by channel and customer segment. Track which actions actually prevent departures and reallocate spend accordingly.	Q2
Low	Credit score has minimal impact on churn (23% vs 19%)	Remove credit score as a retention trigger variable. The 4-point spread does not justify using it to prioritise outreach.	Quarterly

Note on the risk model: The five risk factors used (inactive status, single product, age 40+, balance above \$100K, Germany) were each individually validated against actual churn rates before being included. The model does not require ML — it is a rule-based score that any analyst can run, explain to a business stakeholder, and update when new churn drivers are identified.

8. Conclusion

2,037 customers left out of 10,000 at a 20.37% churn rate. The overall number masks what is happening in specific segments. Germany at 32.4% is running a separate problem from Spain and France. The 50–59 age band at 56% is the highest-risk demographic in the dataset. Female customers churn at 8.6 percentage points above male.

The product count finding is the most counterintuitive result. Two-product customers churn at 7.6% — the bank's most stable segment. Three and four-product customers churn at 83–100%, which means the standard assumption that more products equals more loyalty is wrong in this data. These customers are already leaving.

The risk model identifies 682 HIGH-risk retained customers holding \$89.33M. These are people who match every characteristic of those who already churned. Contacting that list, in balance order, is where this analysis has its most direct return. It does not require a machine learning model, a new data pipeline, or a multi-quarter project. It requires running one query and picking up the phone.

Final Summary Table

Metric	Result
Customers analysed	10,000
Active / Churned	7,963 / 2,037
Overall churn rate	20.37%
Highest-churn country	Germany at 32.4%
Spain / France churn	16.7% / 16.2%
Highest-churn age band	50–59 at 56%
Lowest-churn age band	Under 30 at 8%
Female vs male churn rate	25.07% vs 16.46%
Avg balance — churned vs retained	~\$91,000 vs ~\$73,000
2-product customers (safest)	7.6% churn rate
3-product customers	82.71% churn rate
4-product customers	100% churn rate
Inactive vs active churn rate	~26.85% vs ~14.27%
HIGH-risk retained customers	682 customers
Total balance at risk	\$89.33M
Avg balance per HIGH-risk customer	~\$130,988
Model validation: churned avg risk score	2.67 vs 1.98 (retained)
Recommended immediate action	Run Query 17. Contact top 50 HIGH-risk customers by balance within 7 days.